

# Generator Common Issues and Troubleshooting Methods

## I The generator fails to star.

### 1. The generator does not respond when started by pressing the one-button start switch.

#### 1.1 Check whether the battery can be turned on.

Turn on the battery and the battery level indicators light up.

#### 1.2 Check whether the power cable connector is loosen.

Disconnect the power cable connect and check whether the contact pin is loosen.

**Solution: Fix the contact pin again and attach the cable.**



#### 1.3 Check whether the electric relay is functioning.

Remove the outer shell of the electric relay, and check whether the contact pins of the relay is scorched.

**Solution: Make sure that there is no error with the start motor, then replace the electric relay.**



#### 1.4 Check whether the fuse is abnormal.

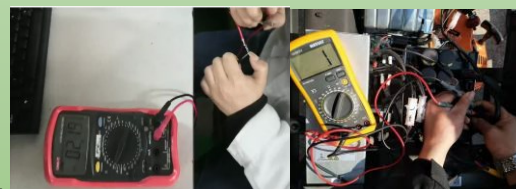
Check whether the fuse is blown.

**Solution: Replace the fuse.**



#### 1.5 Check whether the start motor is functioning.

1. Use the multimeter to check whether the positive and negative poles of the start motor are disconnected.
2. Use a multimeter to check the positive and negative resistance of the start motor, and the resistance should be about 2Ω.
3. Smell with your nose to check whether there is a strong pungent smell in the start motor.



## 2. The generator fails to start when pulling the start handle (or pressing the one-button start switch).

### 2.1 Check whether the engine oil is enough.

Remove the oil dipstick and clean the engine oil on the oil dipstick. Insert the oil dipstick to the crankcase (Not screwed in), then check whether the oil volume is between "L" and "H".

**Solution: Fill enough engine oil.**

## 2.2 Check whether the fuel is sufficient and whether the oil circuit is unobstructed.

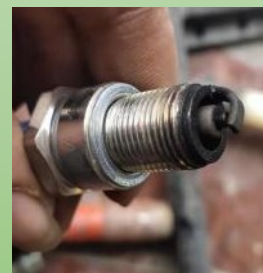
1. Open the fuel tank cap and check whether there is enough 92# gasoline in the fuel tank.
2. Check whether the fuel pipe is misshapen.
3. Turn off the Fuel Switch and loosen the carburetor drain bolt, and drain the fuel in the carburetor. Then turn on the Fuel Switch.
4. Check whether there is gasoline flowing out of the carburetor. Then tighten the carburetor drain bolt.

**Solution: Find out the reason why the oil circuit is obstructed, and then clean or replace the carburetor.**

## 2.3 Check whether the spark plug can ignite.

Remove the spark plug, insert the ceramic end into the ignition cap, and the electrode end to the ground. Pull the starter rope, and check whether the ignition between the spark plug electrodes is normal (obvious spark).

**Solution: Remove the carbon deposits on the electrode. If the issue persists, replace the spark plug.**



## 2.4 The spark plug cannot ignite after using a new spark plug.

1. Check whether the igniter can ignite. Then adjust the multimeter to the resistance position, touch the multimeter probes to the igniter silicon steel sheet and the engine shell. The connection should be normal.
2. Adjust the multimeter to the AC voltage position, touch the multimeter probes to the trigger cable and engine shell, and then pull the starter rope. The voltage should be 0-5V.

**Solution: Replace the igniter or trigger.**

## 2.5 Check whether there is cylinder pressure.

1. Pull the starter rope slowly, and there will be obvious resistance at a certain point (little resistance or no resistance indicates that the valve clearance is small or there is no clearance of the valve; Very large resistance means the valve clearance is large).
2. Remove the spark plug and block the spark plug hole with your fingers. Pull the starter rope and you can feel gas rushing out.

**Solution: Adjust the valve clearance: Inlet valve: 0.10-0.15mm; Exhaust valve: 0.15-0.20mm**

Adjust the valve:

1. Pull the starter rope until there is resistance, and continue to pull it slowly until the pressure is suddenly relieved, which is the top dead center at this time.
2. Remove the spark plug, and insert the long screwdriver to the top of the piston through spark plug hole. Pull the starter rope and when the piston reaches the closest distance to the crankshaft and the two rocker arms are paralleled, it is the top dead center.

## 3. The start handle cannot be pulled.

### 3.1 The engine oil enters the combustion chamber.

1. When filling the engine oil, keep the machine upside down for a long time, making a large amount of engine oil enter the combustion chamber from the exhaust hole.

2. Too much engine oil are filled.

**Solution:** Remove the spark plug, quickly pull the starter rope, and let excess oil drain from the spark plug hole.

### 3.2 There are debris on the magneto.

There are debris on the magneto, causing the stator and rotor unable to rotate smoothly.

**Solution:** Remove the magneto and use a specific tool to remove the rotor and then clean the debris.

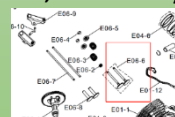
### 3.2 The valve is obstructed or comes off.

Open the cylinder head cover, remove the recoil starter, rotate the starter pulley (clockwise and counterclockwise), and it can rotate smoothly.

If it cannot rotate at a certain point, check whether the rocker arm can push the valve up and down.

**Solution:** If the issue occurs due to carbon deposits, remove the carbon deposits. If the valve comes off, the entire cylinder head will be damaged.

Replace the cylinder head.



### 3.3 There is an error with the crankcase.

1. Drain the engine oil from the crankcase.

2. Remove the muffler.

3. Remove the crankcase cover, rotate the crankshaft, and check whether the crankshaft, connecting rod, camshaft, and balance shaft are abnormal (such as damaged).

**Solution:** Replace the damaged or worn part.

**Tightening torque of screw bolt on the crankcase: 22±2N.m**

## 4. Feel large resistance when pulling the start handle.

### 4.1 The start handle can be pulled, but it feels large resistance at the certain point (rebound).

1. Large valve clearance

**Solution:** Follow the solution mentioned in chapter 2.5 to adjust the clearance.

2. The camshaft pressure relief valve is stuck.

**Solution:** Pull the starter rope to make the piston move to the top dead center, open the crankcase cover and

Check the reason why the camshaft is stuck.

**Align the camshaft to the ignition timing point, mount the camshaft and close the cover, then tighten the screw bolt.**



### 4.2 The rotation speed is slow when the generator starts through the one-button start switch, and it feels large resistance when pulling the start handle.

Disconnect all magneto cables and pull the starter rope to see if there is large resistance.

**Solution:**

Disconnect the cable connecting the motor and inverter, and the starter rope can be pulled normally—Replace the inverter;

There is large resistance after the cable is disconnected—Replace the magneto.

## II Smoke during operation

1. White smoke - There is water in the gasoline
2. Blue smoke (thick smoke) – The engine oil is burnt as too much engine oil are filled.

2.1 Fill too much engine oil - Place the generator horizontally, the engine oil cannot flow out and is stuck between the upper and lower limits of the oil dipstick.

2.2 The generator is seriously tilted during storage and transportation (the cylinder head is facing down).

**Solution:**

**Drain excess engine oil:** Remove the spark plug, and quickly pull the starter rope to let excess engine oil drain from the spark plug hole. Then clean the spark plug and mount it again. Start the generator for 5 to 8 minutes and let the engine oil in the muffler burn out.

2.3 The piston ring is worn – No additional engine oil is filled and the generator is not tilted.

**Solution: Replace the piston ring.**

## III The rotation speed is unstable, the generate shakes greatly, and the charging is slow.

1. The control cable connector is loosen.

1. Check whether the engine's control cable connector comes off.
2. Check whether the contact pin inside the cable connector is loosen.

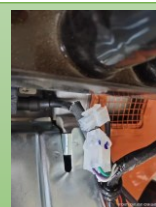
**Solution: Attach the cable connector or contact pin and fix them with the hot melt adhesive.**



2. The connection of the step motor's control cable is poor.

Check whether the cable connector of the step motor comes off.

**Solution: Attach the cable again.**



- 1.3 The step motor is malfunctioning.

The step motor cannot rotate smoothly.

**Solution: Replace the step motor.**



#### 1.4 The choke valve cannot automatically turn on.

1. Check whether the cable connector of the step motor comes off.
2. The step motor is damaged.

**Solution:** Attach the cable again or replace the step motor.

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